

Minh-Anh To

[Email](#) – [LinkedIn](#) – [Portfolio](#)

EDUCATION

Duke University - The Graduate School

Aug 2023 – Exp. May 2025

M.S. - Statistical Science - GPA: 3.91/4.0

Durham, NC

- Full Tuition Scholarship; American Statistical Association Member (ASA).
- Coursework: Predictive Modeling, Deep Learning, Causal Inference, Statistical Computation, R Programming, SQL, Bayesian Statistics, Theory of Inference, Hierarchical Models, Applied Machine Learning.

Hanoi National University of Education

Aug 2014 – May 2018

B.S. - Mathematics (Honors) - GPA: 3.65/4.0 (Top 5%)

Hanoi, Vietnam

National Mathematics Scholarship (Top 1% Applicants); Vice President of Student Council; Dean's List.

SKILLS

Programming: Python (NumPy, pandas, scikit-learn, PyTorch, TensorFlow), R (dplyr, ggplot2, tidy), SQL, Git, REST APIs.

AI/ML: Generative AI (LangChain, OpenAI APIs), Deep Learning (CNNs, RNNs), Machine Learning (XGBoost, Random Forest, Logistic Regression, SVM).

Statistics: Hypothesis Testing, ANOVA, Causal Inference, Forecasting, Bayesian Modeling.

Tools & Platforms: Jupyter, Jekyll, Cloud Services (AWS, GCP).

WORK EXPERIENCE

ChessMind AI

Aug 2024 – Present

AI Developer

Remote, NC

- Developed a **Chess Tutor Chatbot** by utilizing **Generative AI** and collaborating with a **chess grand master** to provide learners with personalized gameplay insights.
- Cleaned and mined large datasets of post-game chess analyses (**100K+ entries**) using **Pandas** and **NLP packages**, building a structured database for training.
- Built a **Retrieval-Augmented Generation (RAG)** pipeline using **LangChain** to retrieve chess concepts from an expert-curated dataset, increasing model **retrieval accuracy** by **28%**.
- Optimized Large Language Model (LLM) responses through **few-shot learning** and **chain-of-thought prompting**, minimizing hallucinations; validated performance with semantic similarity metrics, achieving over **99% accuracy**.
- Documented and maintained a comprehensive GitHub codebase, including unit testing and automation scripts, reducing **pipeline debugging** and **deployment time** by **40%**.

Duke University School of Medicine

May 2024 – Aug 2024

Machine Learning Intern

Durham, NC

- Transformed **10 years** of unstructured antibody production data into a structured database of **5,000 mRNA sequences** and designed an **end-to-end machine-learning pipeline** to predict antibody yields.
- Engineered **35 mRNA structural features** by devising statistical formulas with molecular biologists and coding custom functions in Python, boosting model performance by **75%**.
- Fine-tuned and cross-validated **tree-based models** with scikit-learn, achieving **90% accuracy** and reducing runtime by **48%**; identified 5 key mRNA properties using **feature importance analysis** to enhance yields.
- Authored a detailed report on the model and its **large-scale biotech applications**, leading to selection from 50 candidates to present findings to **200 scientists** at the Duke CFAR Retreat 2024.

Vietnam Academy of Science and Technology

Jun 2022 – Aug 2023

Assistant Researcher

Hanoi, Vietnam

- Demonstrated the practicality of the **bootstrap method** for goodness-of-fit testing using estimator properties; presented findings to **25 researchers** and secured a significant grant from VinIF Vietnam.
- Led a weekly Bayesian Statistics seminar for **30+ students**, including live **R** coding on topics such as **Bayesian Regression**, **Gibbs Sampling**, and **Hierarchical Modeling**.

PROJECT EXPERIENCE

[PeakPlanner - Duke AI Hackathon](#) (Python, LLMs)

[Nuclei Detection with Neural Networks in Breast Cancer Images](#) (Python, Computer Vision, Deep Learning)

[Using Generalized Linear Models for Sparrow Dataset](#) (R, EDA, Statistical Modeling)